Stochastic Modeling and Simulation
3rd Year - Semester 5
2024-2025

Answer Key - Final Exam

Exercice 1 1. *Simulation of a r.v X following the Poisson distribution:*

$P(X=k)=p_k=\frac{\lambda^k}{k!}e^{-\lambda}, k\in\mathbb{N}$ *we can write:* $p_{k+1}=\frac{\lambda}{k+1}p_k$ *with* $p_0=e^{-\lambda}$

Algorithm:

1) Set $k=0, p=e^{-\lambda}, F=p;$

2) Genrate $u\in\mathcal{U}[0,1]$.

3) While $u\geq F$ *set* $p=\frac{\lambda}{k+1}p, F=F+p$ *and* $k=k+1$.

4) Set $X=k$.

(1 pt)

Application: *for* $\lambda=1$,

1)
$$\left|\begin{array}{l} k=0\\ p=e^{-1}=0.3678\\ F=0.3678\\ u=0.0159<F\\ X_1=0 \end{array}\right.$$

(0.25 pt)

2)
$$\left|\begin{array}{l} k=0\\ p=e^{-1}=0.3678\\ F=0.3678\\ u=0.3953>F\Rightarrow\\ \left|\begin{array}{l} p=\frac{\lambda}{1}\times e^{-1}=0.3678\\ F=0.3678+0.3678=0.7356\\ k=1\\ u=0.3953<F\Rightarrow\\ X_2=1 \end{array}\right. \end{array}\right.$$

(0.25 pt)

3)
$$\left|\begin{array}{l} k=0\\ p=e^{-1}=0.3678\\ F=0.3678\\ u=0.8145>F\Rightarrow\\ \left|\begin{array}{l} p=\frac{\lambda}{1}\times 0.3678=0.3678\\ F=0.3678+0.3678=0.7356\\ k=1\\ u=0.8145>F\Rightarrow\\ \left|\begin{array}{l} p=\frac{\lambda}{2}\times 0.3678=0.1839\\ F=0.7356+0.1839=0.9195\\ u=0.8134<F\\ X_3=2 \end{array}\right. \end{array}\right. \end{array}\right.$$

(0.5 pt)

4)
$$\left|\begin{array}{l} k=0\\ p=e^{-1}=0.3678\\ F=0.3678\\ u=0.8397>F\Rightarrow\\ \left|\begin{array}{l} p=\frac{\lambda}{1}\times 0.3678=0.3678\\ F=0.3678+0.3678=0.7356\\ k=1\\ u=0.8397>F\Rightarrow\\ \left|\begin{array}{l} p=\frac{\lambda}{2}\times 0.3678=0.1839\\ F=0.7356+0.1839=0.9195\\ u=0.8397<F\\ X_4=2 \end{array}\right. \end{array}\right. \end{array}\right.$$

(0.5 pt)

The generated sample is $\{0,1,2,2\}$.

2. *Let the random variable X with pdf* $f(x)=\left\{\begin{array}{ll}\frac{1}{4} & \text{if } 0<x<1\\ x-\frac{3}{4} & \text{if } 1\leq x\leq 2\end{array}\right.$.

- *Normalization of f : Let $g(x)$ the pdf of Y, st:* $Y\sim\mathcal{U}[0,2]$, *and* $g_Y(x)=\mathbb{I}_{[0,2]}(x)$, *then*

$$\begin{aligned} C &= \max_{[0,2]}\left|\frac{f(x)}{g(x)}\right| = 2f(2)=\frac{5}{4}\\ &\Leftrightarrow\ 0\leq\frac{f(x)}{Cg(x)}=\frac{4}{5}f(x)\leq 1 \end{aligned}$$

(0.5 pt)

- *Calculate x a value of Y : let $U \in \mathcal{U}[0,1]$ so:* $x = 2u$

Algorithm:

1. *Generate* $U_1, U_2 \in \mathcal{U}[0,1]$,
2. *Calculate* $X = 2U_1$,
3. *Calculate* $\frac{4}{5}f(x) = \begin{cases} \frac{1}{4} & \text{if } 0 < x < 1 \\ \frac{8}{5}u_1 - \frac{3}{5} & \text{if } 1 \leq x \leq 2 \end{cases}$,
4. *If* $u_2 \leq \frac{4}{5}f(x)$ *accept* X, *else reject* X, (1 pt)
5. *Returne to 1.*

Application

App	u_1	$x = 2u_1$	$\frac{4}{5}f(x)$	u_2	*Test*	*Decision*
1	0.0159	$0.0318 \in [0,1]$	0.25	0.3593	$u_2 > \frac{4}{5}f(x)$	*reject* $X = 0.0318$
2	0.8145	$1.629 \in [1,2]$	0.5532	0.8397	$u_2 > \frac{4}{5}f(x)$	*reject* $X = 1.629$
3	0.3090	$0.618 \in [0,1]$	0.25	0.8100	$u_2 > \frac{4}{5}f(x)$	*reject* $X = 0.618$

(1.5 pt)

Exercice 2 1. *Estimation of* $I = \int_0^{\frac{\pi}{2}} \sin(x)\,dx$ *using Monte Carlo method (average sampling)*

a) *We can see that*

$$I = \int_{\mathbb{R}} \frac{\pi}{2}\sin(x) \times \frac{2}{\pi}\mathbb{I}_{[0,\frac{\pi}{2}]}(x)\,dx = \mathbb{E}\left[\frac{\pi}{2}\sin(X)\right],$$

where $g(x) = \frac{\pi}{2}\sin(x)$ *and* $f_X(x) = \frac{2}{\pi}\mathbb{I}_{[0,\frac{\pi}{2}]}(x)$ *is pdf of r.v.* X *following a uniform distribution* $\mathcal{U}\left[0,\frac{\pi}{2}\right]$. *Applying the MCM, we can estimate* I *by*

$$\widehat{I}_N = \frac{1}{N}\sum_{i=1}^{N}\frac{\pi}{2}\sin(X_i),$$ (1 pt)

with $(X_i)_{1\leq i\leq N}$ *i.i.d. following* $\mathcal{U}\left[0,\frac{\pi}{2}\right]$.

b. *Let* $Z_i = \frac{\pi}{2}\sin X_i$; *given that*

$$\begin{aligned}\mathbb{E}\left[Z^2\right] &= \mathbb{E}\left[\frac{\pi^2}{4}\sin^2(x)\right] = \frac{\pi^2}{4}\int_0^{\frac{\pi}{2}}\frac{2}{\pi}\sin^2(x)\,dx \\ &= \frac{\pi}{2}\int_0^{\frac{\pi}{2}}\frac{1-\cos(2x)}{2}dx < +\infty,\end{aligned}$$

then, we can calculate the variance of $\widehat{I}_N$.

$$Var\left(\widehat{I}_N\right) = \frac{Var(Z)}{N}.$$

Remak that $I = [-\cos x]_{x=0}^{x=\frac{\pi}{2}} = 1$. *On the other hand,*

$$\begin{aligned}Var(Z) &= \mathbb{E}\left[Z^2\right] - I^2 = \int_0^{\frac{\pi}{2}}\frac{\pi^2}{4}\left(\frac{1}{2} - \frac{\cos(2x)}{2}\right)\frac{2}{\pi}dx - 1 \\ Var(Z) &= \frac{\pi}{2}\left[\frac{1}{2}x - \frac{1}{4}\sin(2x)\right]_0^{\frac{\pi}{2}} - 1 = \frac{\pi^2}{8} - 1,\end{aligned}$$ (1 pt)

hence

$$Var\left(\widehat{I}_N\right) = \frac{1}{N}\left(\frac{\pi^2}{8} - 1\right). \tag{1}$$

2. *Estimate* $I = \int\limits_{\mathbb{R}} g(x) f_X(x)\,dx = \int_0^{\frac{\pi}{2}} \frac{\pi}{2}\sin(x) \times \frac{2}{\pi}dx$, *using the importance sampling method*

   a) *We approximate* $\sin(x)$ *by an odd polynomial* x *of degree* 1 *equal to* 0 *for* $x = 0$ *and* $\frac{\pi}{2}$ *for* $x = \frac{\pi}{2}$. *So we define* $\widetilde{f} \sim |g \times f|$ *by* $\widetilde{f}(x) = \lambda x \times \frac{2}{\pi}\mathbb{I}_{\left[0,\frac{\pi}{2}\right]}(x)$, *then:*

$$\int \widetilde{f}(x)\,dx = 1 \Longleftrightarrow \frac{2\lambda}{\pi}\int_0^{\frac{\pi}{2}} x\,dx = 1 \Longleftrightarrow \lambda = \frac{8}{\pi^2}$$

hence, $\widetilde{f}(x) = \frac{8}{\pi^2}x\mathbb{I}_{\left[0,\frac{\pi}{2}\right]}(x)$ *that is pdf of r.v.* Y, *then*

$$\begin{aligned} I &= \int\limits_{\mathbb{R}} \frac{g(x) f(x)}{\widetilde{f_Y}(x)}\widetilde{f_Y}(x)\,dx = \int\limits_{\mathbb{R}} \frac{\frac{\pi}{2}\sin(x) \times \frac{2}{\pi}\mathbb{I}_{\left[0,\frac{\pi}{2}\right]}(x)}{\frac{8}{\pi^2}x} \times \frac{8}{\pi^2}x\,dx \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{\pi^2}{8}\frac{\sin(x)}{x}\right)\frac{8}{\pi^2}x\,dx = \mathbb{E}\left[\frac{\pi^2}{8}\frac{\sin(Y)}{Y}\right], \end{aligned}$$

so we can approximate I *by*

$$I \approx \widehat{M}_N = \frac{1}{N}\sum_{i=1}^{n}\frac{g(X_i) f(X_i)}{\widetilde{f}(X_i)} = \frac{1}{N}\sum_{i=1}^{n}\frac{\pi^2}{8}\frac{\sin(Y_i)}{y_i},$$

(1 pt)

with $(Y_i)_{i=1}^n$ *i.i.d of pdf* $\widetilde{f}$.

b) Let $W = \frac{\pi^2}{8}\frac{\sin(Y)}{Y}$, we have

$$\begin{aligned} \mathbb{E}\left[W^2\right] &= \frac{\pi^4}{8^2}\int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{x^2}\frac{8}{\pi^2}x\,dx \\ &= \frac{\pi^2}{8}\int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{x}dx \le \frac{\pi^2}{8}\int_0^{\frac{\pi}{2}} \frac{x}{x}dx < +\infty, \end{aligned}$$

then, we can calculate the variance of $\widehat{M}_N$.

$$Var\left(\widehat{M}_N\right) = \frac{Var(W)}{N}.$$

$$Var(W) = \mathbb{E}\left[W^2\right] - \mathbb{E}[W]^2 \le \frac{\pi^2}{8}\int_0^{\frac{\pi}{2}} \frac{x}{x}dx - 1 \le \frac{\pi^3}{16} - 1,$$

so,

$$Var\left(\widehat{M}_N\right) = \frac{1}{N}\left(\frac{\pi^3}{16} - 1\right). \tag{2}$$

(1 pt)

3.

a) Comparaison of estimators, from (1) and (2) we have

$$N\left[Var\left(\widehat{I}_N\right) - Var\left(\widehat{M}_N\right)\right] = \frac{\pi^2}{8} - \frac{\pi^3}{16} = \frac{2\pi^2 - \pi^3}{16} < 0 \qquad \text{(0.5 pt)}$$

so the importance sampling does not reduce variance then, $\widehat{I}_N$ is a best estimator of I.

b) Give an approximate number of iterations required to achieve an accuracy of the previous estimator ± 0.01.

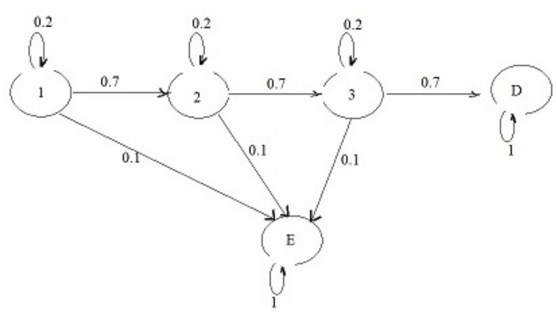
We have $\frac{\sigma}{\sqrt{N}} = \epsilon_N \leq 0.01$ and, since $\sigma^2 = Var\left(\widehat{I}_N\right) = \frac{1}{N}\left(\frac{\pi^2}{8} - 1\right)$ we obtain

$$\begin{aligned} \frac{\sigma^2}{N} &\leq 0.01 \Leftrightarrow \frac{1}{N^2}\left(\frac{\pi^2}{8} - 1\right) \leq 0.0001 \\ &\Leftrightarrow N^2 \geq \left(\frac{\pi^2}{8} - 1\right) \times 10^4 = 0.23370055 \times 10^4 \\ &\Leftrightarrow N \geq 48.35 \end{aligned}$$

So, for $N = 49$ we find $Var\left(\widehat{I}_N\right) = \frac{1}{49}\left(\frac{\pi^2}{8} - 1\right) \simeq 0.0048$, acceptable. (1 pt)

Exercice 3 *In higher education institution*

1. *The transition graph associated to the Markov chain with states space* $S = \{1, 2, 3, E, D\}$ *and, two absorbant state* E *and* D



(0.5 pt)

And its transition matrix

$$P = \begin{pmatrix} 0.2 & 0.7 & 0 & 0.1 & 0 \\ 0 & 0.2 & 0.7 & 0.1 & 0 \\ 0 & 0 & 0.2 & 0.1 & 0.7 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \qquad \text{(0.5 pt)}$$

2. *The probabilities that a student obtains his diploma based on his current state in the cursus are:*

$$\begin{aligned} p_{1D} &= 0.2p_{1D/11} + 0.7p_{2D} \\ p_{2D} &= 0.2p_{2D/22} + 0.7p_{3D} \\ p_{3D} &= 0.2p_{3D/33} + 0.7p_{DD} \end{aligned}$$

since $p_{DD} = 1$ *(absorbant state), the previous system becomes*

$$\begin{aligned} p_{3D} &= 0.2 p_{3D/33} + 0.7 \\ p_{2D} &= 0.2 p_{2D/22} + 0.7 \times 0.2 p_{3D/33} + (0.7)^2 \\ p_{1D} &= 0.2 p_{1D/11} + 0.7 \times 0.2 p_{2D/22} + (0.7)^2 \times 0.2 p_{3D/33} + (0.7)^3 \end{aligned}$$

then the solution of this system is

$$\begin{aligned} p_{3D} &= 0.2 \times 0.7 + 0.7 = 0.84, \\ p_{2D} &= 0.2 \times (0.7)^2 + 0.7 \times 0.84 = 0.686, \\ p_{1D} &= 0.2 \times (0.7)^3 + 0.7 \times 0.686 = 0.5488. \end{aligned}$$ (1.5 pt)

3. *The average duration of studies: the end of studies is associated with states* E *and* D*; we group them into a state* F *(finish or end of studies) and then consider the chain with four states.*

   Let $\overline{T_i}, i \in \{1,2,3\}$ *the the average stay time in the transient states* $\{1,2,3\}$ *before absorption by* F*, so the average duration of studies is* $\overline{T_1}$ *and we have*

$$\begin{aligned} \overline{T_1} &= 1 + 0.2\overline{T_1} + 0.7\overline{T_2} \\ \overline{T_2} &= 1 + 0.2\overline{T_2} + 0.7\overline{T_3} \\ \overline{T_3} &= 1 + 0.2\overline{T_3} \end{aligned}$$

   the system solution is: $\overline{T_3} = \frac{1}{1-0.2} = \frac{1}{0.8}$ *and* $\overline{T_2} = \frac{0.8+0.7}{(0.8)^2} = \frac{1.5}{0.64}$*, so:*

$$\overline{T_1} = \frac{1 + 0.7\overline{T_2}}{0.8} = \frac{0.64 + 0.7 \times 1.5}{(0.8)^3} = \frac{1,69}{0.512} = 3.3 \ years.$$ (1.5 pt)

4. ***Algorithm:***

   Initial state $X_0 = 1$

   1) Generate u_1*, if* $u_1 < 0.2 \rightarrow X_1 = 1$

   if $0.2 \leq u_1 < 0.9 \rightarrow X_1 = 2$

   if $0.9 \leq u_1 < 1 \rightarrow X_1 = E$

   2) If $X_1 = 2$*, generate* u_2*, if* $u_2 < 0.2 \rightarrow X_2 = 2$

   if $0.2 \leq u_2 < 0.9 \rightarrow X_2 = 3$

   if $0.9 \leq u_2 < 1 \rightarrow X_2 = E$

   3) If $X_2 = 3$*, generate* u_3*, if* $u_3 < 0.2 \rightarrow X_3 = 3$

   if $0.2 \leq u_3 < 0.9 \rightarrow X_3 = D$

   if $0.9 \leq u_3 < 1 \rightarrow X_3 = E$

   4) In step n*, if* $X_{n+1} = X_n \rightarrow$ *return to step* n*,*

   if $X_{n+1} = D$ *stop.*

   if $X_{n+1} = E$ *stop.* (1.5 pt)

Dr N. ARRAR

Exercice 4 *In service company*

Model A *Queueing system $M/M/1$, with arrival rate λ and service rate 2μ.*

Since $\frac{\lambda}{2\mu} < 1$, the system is in stationary state, we have:

$$\overline{n} = \frac{\rho}{1-\rho} = \frac{\lambda}{2\mu\left(1 - \frac{\lambda}{2\mu}\right)} = \frac{\lambda}{2\mu - \lambda} \qquad \text{(0.5 pt)}$$

So, from the Little' law the average time spent in the system is

$$\overline{W_{S_A}} = \frac{\overline{n}}{\lambda} = \frac{1}{2\mu - \lambda}. \qquad \text{(0.5 pt)} \tag{3}$$

Model B *The sum of two identical systems $M/M/1$, In each one the arrival rate is $\frac{\lambda}{2}$ and the service rate is μ.* (0.5 pt)

So, the average time spent in the system is

$$\overline{W_{S_B}} = \frac{1}{2}\frac{1}{\mu - \frac{\lambda}{2}} + \frac{1}{2}\frac{1}{\mu - \frac{\lambda}{2}} = \frac{2}{2\mu - \lambda}. \qquad \text{(0.5 pt)} \tag{4}$$

Model C *Queueing system $M/M/2$, with arrival rate λ and service rate is μ per server.*

We have $s = 2$,

$$\begin{aligned} p_0 &= \left[1 + \frac{\lambda}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^2}{2\left(1 - \frac{\lambda}{2\mu}\right)}\right]^{-1} = \left[1 + \frac{\lambda}{\mu} + \frac{\lambda^2}{\mu\,(2\mu - \lambda)}\right]^{-1} \\ &= \left[\frac{2\mu^2 - \lambda\mu + 2\lambda\mu - \lambda^2 + \lambda^2}{\mu\,(2\mu - \lambda)}\right]^{-1} = \frac{2\mu - \lambda}{2\mu + \lambda}. \qquad \text{(1 pt)} \end{aligned}$$

So,

$$\begin{aligned} \overline{W_{S_C}} &= \frac{1}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^s}{s\mu s!\left(1 - \frac{\lambda}{s\mu}\right)^2} p_0 = \frac{1}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^2}{4\mu\left(\frac{2\mu-\lambda}{2\mu}\right)^2} \times \frac{2\mu - \lambda}{2\mu + \lambda} \\ &= \frac{1}{\mu} + \frac{\lambda^2}{\mu\,(2\mu - \lambda)^2} \times \frac{2\mu - \lambda}{2\mu + \lambda} = \frac{4\mu^2 - \lambda^2 + \lambda^2}{\mu\,(2\mu - \lambda)\,(2\mu + \lambda)} \\ \overline{W_{S_C}} &= \frac{4\mu}{(2\mu - \lambda)\,(2\mu + \lambda)}. \qquad \text{(1 pt)} \end{aligned} \tag{5}$$

Comparaison: *Since $\lambda < 2\mu$ then $2\mu + \lambda < 4\mu$, so: $\frac{4\mu}{2\mu+\lambda} > 1$ and $\overline{W_{S_C}} > \overline{W_{S_A}}$.* (0.5 pt)
In the other hand, $\overline{W_{S_B}} > \overline{W_{S_A}}$.
From (4) and (5)

$$\frac{4\mu}{(2\mu + \lambda)} < \frac{4\mu}{2\mu} = 2 \ so, \overline{W_{S_B}} > \overline{W_{S_C}}. \qquad \text{(0.5 pt)}$$

Then

$$\overline{W_{S_A}} < \overline{W_{S_C}} < \overline{W_{S_B}}.$$

Model A is the best performing. (0.5 pt)